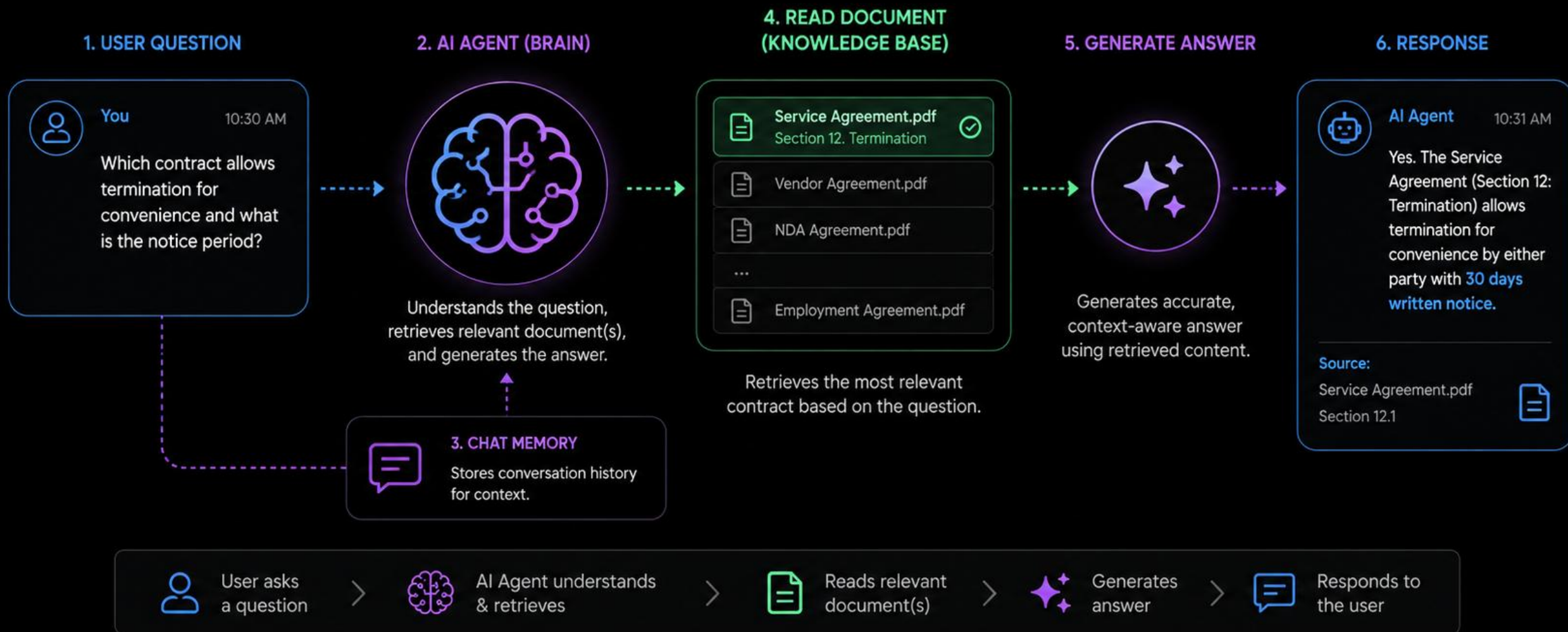


RAG in Action

Retrieve relevant information. Generate accurate answers.



RAG combines the power of retrieval and generation to provide accurate and trustworthy answers.

THE PROBLEM: FINDING ONE ANSWER IN 1,000 DOCUMENTS

Can an AI read **everything** every time?

1. USER QUESTION

 You 10:30 AM

Which vendor agreements allow termination for convenience with 30 days' notice?

Type your question... 

2. AI AGENT (OVERWHELMED)



3. THE REALITY: 1,000+ DOCUMENTS



WITHOUT RETRIEVAL (READING EVERYTHING)



 Reading everything is NOT scalable, NOT affordable, and NOT reliable.



HOW DOES THE AI KNOW WHICH DOCUMENTS TO READ?

There has to be a smarter way...

THE SOLUTION: INDEX FIRST, ANSWER SMART

Organize → Index → Retrieve → Answer. Read less. Know more.



BEFORE: WITHOUT INDEX (READ EVERYTHING)



Problems

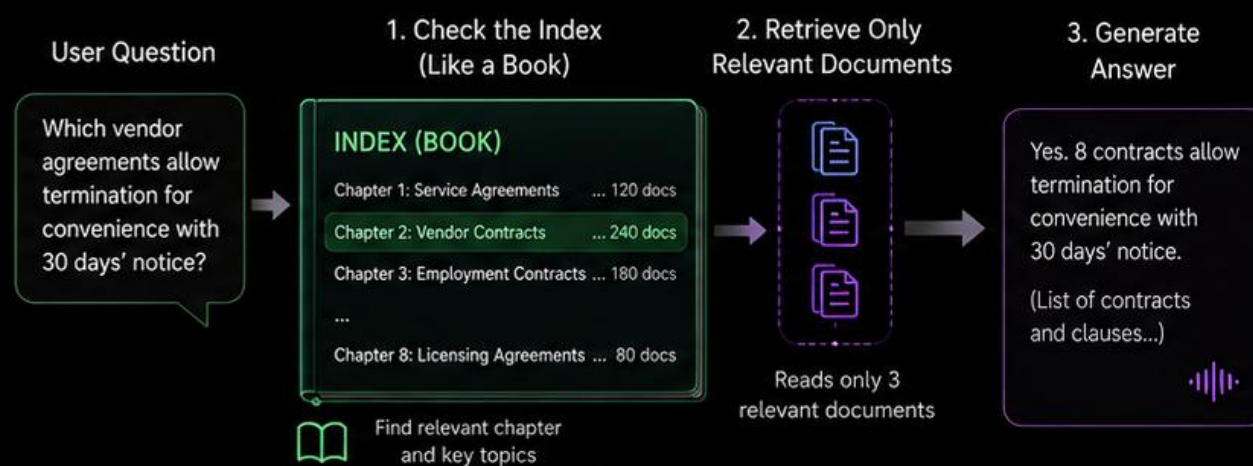
- ⊗ Reads all documents every time
- ⊗ Millions of input tokens
- ⊗ Very high cost
- ⊗ Slow responses
- ⊗ Hard to scale
- ⊗ Higher risk of missing key info

ESTIMATED COST PER QUESTION

Input Tokens: 2,500,000+
Output Tokens: 2,000+
Est. Cost: \$0.80 – \$1.20+



AFTER: WITH INDEX (RAG APPROACH)



Benefits

- ✓ Reads only what's relevant
- ✓ Dramatically fewer tokens
- ✓ Much lower cost
- ✓ Faster responses
- ✓ Scales to thousands of documents
- ✓ More accurate answers

ESTIMATED COST PER QUESTION

Input Tokens: 8,000 – 15,000
Output Tokens: 1,500 – 2,500
Est. Cost: \$0.003 – \$0.01

COST SAVINGS

90–99%
LOWER COST



KEY IDEA

Don't ask the LLM to read everything.
Guide it to the **right information first.**

Metric	Without Index (Read Everything)	With Index (RAG Approach)	Improvement
Documents Read	1,000+	3 – 5 relevant docs	99% fewer documents read
Input Tokens (per question)	2,500,000+	8,000 – 15,000	99% fewer input tokens
Output Tokens (per question)	2,000+	1,500 – 2,500	Similar / Slightly lower
Estimated Cost (per question)	\$0.80 – \$1.20+	\$0.003 – \$0.01	90–99% lower cost
Speed	Slow 🐢	Fast ⚡	10x – 50x faster
Scalability	Does NOT scale ⊗	✓ Scales to 10K+ documents	Enterprise ready